

Self-Supervised Anchoring of Fingertip Sensing to Proprioception and Proactive Actions for Robot Imitation Learning

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Abstract—Robotic imitation learning often relies on external cameras, yet local interaction cues such as object proximity, contact onset, and grasp state are difficult to observe near the fingertips because of occlusion and limited temporal resolution. We study how to effectively incorporate complementary fingertip sensing into imitation learning using pressure-sensitive tactile and reflective proximity sensors, along with pretrained sensor encoders. The two modalities provide information at different manipulation phases: proximity sensing is informative before contact, whereas tactile sensing becomes informative after contact. However, naively adding these signals to a policy does not consistently improve performance and can even underperform vision-only policies, suggesting that sparse, phase-dependent sensor signals are difficult to exploit from limited demonstrations. We therefore propose a proprioception-anchored pretraining method, **PRO**prioceptive-and-**PRO**active Anchoring (PROPRA), which independently aligns each fingertip sensor history with proprioceptive and action segments. This provides a continuously available sensorimotor reference, allowing each sensor to be aligned independently during its informative phases. Experiments on real-world manipulation tasks show that our pretraining method improves average success rates over vision-only policies and image-anchored pretraining baselines. Representation analysis further shows that it preserves richer information about pre-contact states, enabling more effective use of complementary fingertip sensing. Please refer to our project page: <https://anonymous.4open.science/w/own-F865/>.

I. INTRODUCTION

Fingertip sensing provides rich perceptual information that plays a central role in dexterous human manipulation. Humans flexibly adapt their movements to the characteristics and state of a task by integrating multiple sensory signals, including vision and touch. Even in fundamental grasping behaviors, such multimodal feedback enables us to execute precise movements and to maintain reliable grasping even when visual information is partially or temporarily unavailable. Endowing robotic systems with similar fingertip sensing capabilities is therefore a promising direction toward more dexterous and robust manipulation.

The importance of incorporating multimodal perceptual information near the fingertips into imitation learning for robotic manipulation has been increasingly recognized [1]–[3]. However, different sensors provide distinct types of information, and their informativeness varies across the manipulation process. In particular, recognizing not only the post-contact state but also the *pre-grasp* state, in which the robot approaches the object and establishes a graspable spatial configuration, is known to be crucial for determining the subsequent contact and grasp outcome [4]–[6]. Yet, even when complementary sensing is exploited, naively concatenating

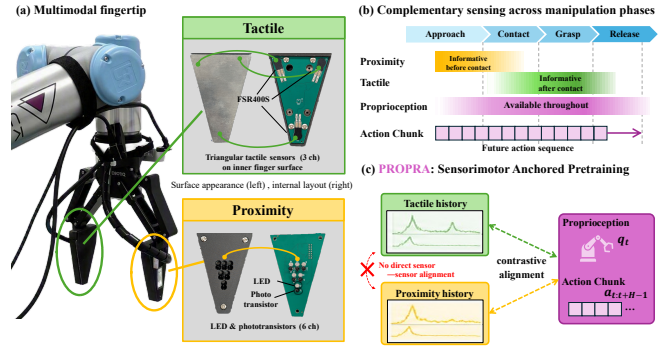


Fig. 1: Overview of the proposed multimodal fingertip and PROPRA. (a) Multimodal fingertip with tactile and proximity sensors. (b) Complementary sensing across manipulation phases: proximity before contact, tactile after contact, with proprioception available throughout and future action chunks providing additional sensorimotor context. (c) PROPRA independently aligns tactile and proximity histories.

modalities can degrade the policy on some tasks, where sensor-competition effects interfere with each other [7]. Importantly, the challenge extends beyond policy fusion. When sensory modalities become informative at different phases, learning representations by directly aligning modalities can also produce weak or ambiguous supervision signals. For example, a modality may provide little or no informative signal during periods when another modality is most relevant, making temporal co-occurrence an unreliable supervision cue. Because each modality becomes informative at different stages of manipulation, the learned representation should capture when each sensor contains task-relevant information and how that information relates to the robot’s motion and physical state. Representation learning is therefore important for converting heterogeneous fingertip signals into features that an imitation learning policy can effectively exploit.

To address this challenge, we investigate tactile and proximity sensing, which exhibit heterogeneous informative phases for imitation learning, and propose **PRO**prioceptive-and-**PRO**active Anchoring (PROPRA). This contrastive pre-training framework uses proprioception and action sequences, available throughout manipulation, as a common reference for learning fingertip sensor representations, where *proactive* refers to action chunks that encode the robot’s intended motion. By combining fingertip histories, current proprioception, and proactive actions, PROPRA temporally links fingertip perception with the robot’s embodied state across past, present, and near-future interaction. Unlike image-

anchored approaches, which rely on visual observations at a single time step, PROPRA uses proprioception and proactive actions that remain available throughout manipulation as anchors. This avoids forcing modalities that respond in different phases, while associating each modality with the states and motion during its informative phase.

The main contributions of this work are threefold:

- We develop a fingertip sensing system that combines pre-contact proximity sensing with post-contact tactile sensing, and demonstrate its ability to capture complementary interaction states throughout the grasping process in real-robot experiments.
- We formulate the representation-learning challenge for complementary fingertip sensors that provide informative signals at different manipulation phases, and propose PROPRA, which uses proprioception and action chunks as a sensorimotor anchor.
- We demonstrate higher average success rates than vision-only policies and image-anchored pretraining, while retaining pre-contact information in the learned representation, including on recordings from policy execution.

II. RELATED WORK

Fingertip sensing provides direct information about physical interaction without relying solely on visual inference. Fundamental skills such as grasping and insertion particularly benefit from local perception, since vision-based position control can be unreliable when accurate interaction states must be estimated. We therefore review prior work on fingertip sensing and representation learning for multiple sensing modalities.

Tactile sensing has been widely used to improve robotic dexterity by directly measuring robot–object interaction [1]. Vision-based tactile sensors observe contact-induced deformation of an elastic surface and are well suited to describing post-contact states such as object geometry, contact force, slip, and object pose during grasping [8]–[10]. Alongside recent learning-based manipulation systems that generate actions from camera observations [11], prior work has explored self-supervised visuotactile learning [12], shared representations across optical tactile sensors [13], and patch-level visual–tactile alignment [14].

However, tactile sensing alone cannot directly observe the interaction state before physical contact. Proximity sensing has therefore been used to estimate fingertip–object distance and relative configuration during the *pre-grasp* or *pre-shaping* phase [2], [4], [5]. Proximity sensing is informative before contact, whereas tactile sensing becomes informative after contact. Combining them thus provides complementary observations over a broader range of the grasping process.

This complementary sensing is also relevant to Learning from Demonstration (LfD), where fingertip information has been shown to improve manipulation learning [2], [12], [15]. A key challenge is how to learn useful sensor representations from limited demonstrations. Contrastive pretraining provides one approach by exploiting relationships among observations to structure the latent space. Visual representation

learning has used temporal relationships and goal states in demonstrations [16], [17]. For tactile sensing, exUMI [18] predicts future tactile observations conditioned on future actions, while τ [19] uses future visual observations as supervision. Visuotactile pretraining can also improve downstream policy performance [3]. TactX [20] learns sensor-independent representations across optical, magnetic, and resistive tactile sensors, while HTT [21] aligns optical and low-dimensional tactile observations through masked autoencoding and cross-modal alignment.

When multiple modalities must be represented jointly, a common strategy is to align them through a shared anchor. ImageBind [22] uses images as an anchor and independently aligns other modalities to a shared latent space, while Uni-Touch [23] applies a similar strategy to tactile representation learning. CentroBind [24] further points out limitations of a fixed anchor modality, including limited anchor information, loss of modality-specific information, and insufficient modeling of correlations among non-anchor modalities.

Fingertip sensors are informative at different interaction phases, so direct same-time alignment may provide insufficient supervision. PROPRA instead uses proprioception and action sequences as a common sensorimotor anchor to learn complementary *pre-grasp* and contact-related representations.

III. PROBLEM FORMULATION

We first examine how tactile and proximity sensing become informative at different stages of manipulation and clarify the resulting challenge for representation learning. By combining proximity sensing, which responds before contact, with tactile sensing, which becomes informative after contact, on the same fingertip, the interaction can be observed continuously from approach through contact to object holding. Proximity sensing captures the fingertip–object distance and the progression toward contact during approach, while tactile sensing captures contact onset and the subsequent grasp state. Based on this complementary sensing principle, we developed a fingertip module that integrates tactile and proximity sensors on the same fingertip (Fig. 1).

A. Hardware design

For tactile sensing, we use force-sensitive resistors (FSR400S). For proximity sensing, we use a reflective optical distance-sensing circuit. All sensor signals are acquired at 100 Hz and stored in synchronization with the robot trajectory. Both sensing elements are commercially available components and can be retrofitted to the fingertips of an existing parallel gripper.

B. Interval Structure of Sensor Responses

Let s_t^m denote the observation of modality $m \in \text{tac, prox}$ at time t , with q_t denoting the robot state and a_t the action. We define the contact onset t_c as the time at which the tactile signal rises from its stationary baseline; the detection criterion is described in Section V-C. The interval before t_c is defined as *pre-grasp*. Within this interval, the 60 control steps

TABLE I: Effect of naively adding fingertip sensing. Success rate [%], mean $\pm$ SD over 3 seeds with 50 trials per seed. Sensor histories are concatenated to the observation without pretraining.

Task	Vision only	+ Tactile & proximity	Δ
PickCup	11.7 $\pm$ 7.6	61.7 $\pm$ 5.8	+50.0
PickSponge	35.0 $\pm$ 5.0	51.7 $\pm$ 12.6	+16.7
MovePen	31.7 $\pm$ 10.4	20.0 $\pm$ 8.7	-11.7
OpenLid	43.3 $\pm$ 7.6	28.3 $\pm$ 15.3	-15.0
Avg.	30.4	40.4	+10.0

(approximately 2 s) immediately preceding t_c are defined as *approach*, and the earlier interval as *pre*. The interval from t_c onward is defined as *contact*. These phase labels are used only for analysis and evaluation and are not provided during training.

The characteristics of the two sensor signals change across these phases. During *pre*, both sensors remain near their stationary baselines. During *approach*, the proximity signal begins to vary and becomes most discriminative immediately before contact, while the tactile signal remains near baseline. During *contact*, the tactile signal rises in response to physical interaction. Thus, proximity sensing mainly covers the pre-contact interval, whereas tactile sensing becomes informative after contact. However, each sensor is discriminative only over a limited portion of the episode. In the demonstration data, at least one of the two sensors remains near its stationary value for more than half of the pretraining samples (Section V-C).

Having access to sensor information does not necessarily mean that a policy can use it effectively. When sensor histories are directly concatenated to a diffusion policy [25] that otherwise uses images and robot state, performance falls below the vision-only policy on some tasks (Table I). With limited demonstrations, the policy must learn both the meaning of each sensor signal and the interaction phase in which that signal should be used. This motivates learning sensor representations before policy training.

C. Missing Supervision in Contrastive Pretraining

We next consider contrastive pretraining that aligns observations acquired at the same time as positive pairs. If one element of a pair remains constant over time, the loss does not require the encoder of the other element to preserve temporal distinctions. Because the informative intervals of tactile and proximity sensing only partially overlap, direct alignment yields supervision only in a limited portion of the trajectory. During *approach*, the tactile signal remains near its stationary baseline even though the proximity signal becomes discriminative, so direct tactile-proximity alignment does not require the proximity encoder to distinguish states within this interval. Indeed, tactile observations cannot predict the corresponding proximity variation during *approach* in our demonstration data (Section V-C). Removing degenerate samples does not resolve this issue: the remaining direct

pairs are still concentrated near the contact transition, leaving most of the *approach* phase without useful supervision.

From this observation, we require an anchor to satisfy the following three conditions:

- (i) It is defined throughout manipulation, including intervals in which individual sensors do not respond.
- (ii) It changes with the progress of manipulation and systematically covaries with the response intervals of the sensors.
- (iii) It contains complementary sensorimotor context that is not a deterministic copy of any single sensor observation.

Condition (i) ensures that a common reference is available throughout the manipulation sequence, while useful supervision for each sensor remains concentrated in the phases where that sensor is informative; condition (ii) provides a meaningful relationship between the anchor and the interaction state, and condition (iii) prevents the anchor from reducing to an identity-like alignment with a single sensing modality. Tactile and proximity observations do not satisfy condition (i) when used as anchors for one another. Images satisfy condition (i), but their relationship to local physical quantities at the fingertips is indirect; whether condition (ii) holds depends on how clearly the fingertip-object interaction is observable in the image.

We therefore seek an anchor that is available throughout manipulation while remaining strongly coupled to the robot interaction state and motion. The next section introduces the sensorimotor anchor adopted in PROPRA to satisfy these requirements.

IV. METHOD

A. Proprioceptive and Proactive Anchored Pretraining

PROPRA uses proprioception and action sequences as a common sensorimotor anchor for fingertip sensors whose informative signals arise at different stages of interaction, learns complementary representations of *pre-grasp* and contact-related states, and transfers them to policy learning. It has two stages: pretraining of the sensor encoders (Fig. 2) and policy learning (Fig. 3). In pretraining, a modality-specific encoder maps each sensor history to an embedding that is aligned independently with a common anchor embedding; sensor modalities are never paired directly, and their correspondence is formed only through the shared anchor. In policy learning, only the pretrained sensor encoders are transferred as observation encoders of the diffusion policy; the anchor and its encoder are used solely in pretraining.

For each sensor encoder, we use the most recent L -step history ending at time t . The anchor consists of the proprioceptive state at time t and an H -step action chunk starting from t :

$$x_t^m = [s_{t-L+1}^m, \dots, s_t^m] \in \mathbb{R}^{L \times d_m}, \quad (1)$$

$$\bar{a}_t = [a_t^\top, a_{t+1}^\top, \dots, a_{t+H-1}^\top]^\top \in \mathbb{R}^{HA}, \quad (2)$$

$$h_t = [q_t^\top, \bar{a}_t^\top]^\top \in \mathbb{R}^{P+HA}. \quad (3)$$

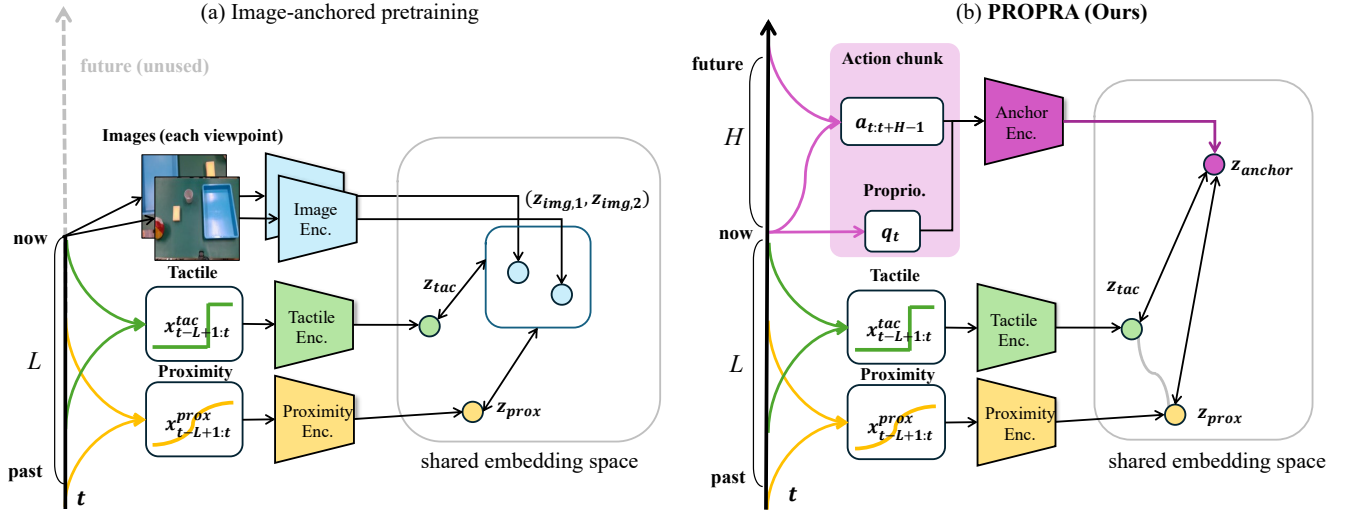


Fig. 2: **Pre-training for sparse-response sensors.** (a) Image-anchored alignment: sensor histories (past L steps) are aligned to image embeddings at time t ; effective supervision concentrates in the brief contact window. (b) Our **PROPRA** aligns each sensor history to a temporally shared reference, the current state q_t , and the forthcoming action chunk $a(t : t + H - 1)$, which is defined and varies at every t ; no images are used. Dashed line in (b): no direct sensor-to-sensor objective — cross-sensor alignment emerges through the shared anchor.

In our implementation, q_t includes the end-effector pose and gripper opening ($P = 8$), while a_t includes seven commanded joint positions and one gripper command ($A = 8$). With $H = 10$, the anchor h_t is therefore 88-dimensional.

The anchor h_t satisfies the three requirements of Section III-C: it is defined over the entire trajectory because the robot always has a proprioceptive state and receives control commands (i); it covaries with the sensors’ response phases, since the end effector moves toward the object before contact and the gripper command changes as the fingers close around contact (ii); and it is built entirely from robot-side variables, so it is not a deterministic copy of either tactile or proximity observations (iii).

The windows in Eqs. (1) and (2) are asymmetric around t : the sensor input covers the history up to t , whereas the anchor covers the current proprioceptive state and the actions from t onward. PROPRA thus associates each sensor history with the current embodied state and subsequent motion rather than matching simultaneous observations, a predictive structure related to Contrastive Predictive Coding [26]. The positive pairs at each t are (h_t, x_t^m) for each modality m ; no direct pair (x_t^{tac}, x_t^{prox}) is used, and each additional modality adds a single anchor–modality term, so the framework scales linearly with the number of sensors.

B. Encoders and Learning Objective

Each modality with d_m channels is normalized channel-wise by its mean and standard deviation; action chunks extending beyond an episode are padded by repeating the final action. Each sensor encoder g_m is a four-layer pre-norm Transformer encoder: inputs are linearly projected to $d_{model} = 128$ with sinusoidal positional encoding, mean-pooled over time, and projected to a 512-dimensional embedding. Encoder parameters are not shared, because the

modalities differ in physical quantity and response characteristics. The anchor encoder g_a is a multilayer perceptron. All embeddings are L_2 -normalized, $u_i^m = g_m(x_i^m)/\|g_m(x_i^m)\|$ and $z_i = g_a(h_i)/\|g_a(h_i)\|$, so that their inner product corresponds to cosine similarity.

Within a mini-batch of B samples, z_i and u_i^m from the same time step form a positive pair, and the remaining u_j^m ($j \neq i$) are negatives. For each modality, we use the symmetric InfoNCE objective [26], [27], averaged over both directions:

$$\mathcal{L} = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} \mathcal{L}_m, \quad (4)$$

$$\mathcal{L}_m = -\frac{1}{2B} \sum_{i=1}^B \left[\log \frac{\exp(z_i^\top u_i^m / \tau)}{\sum_{j=1}^B \exp(z_i^\top u_j^m / \tau)} + \log \frac{\exp(z_i^\top u_i^m / \tau)}{\sum_{j=1}^B \exp(z_j^\top u_i^m / \tau)} \right]. \quad (5)$$

where τ is a learnable temperature. The tactile and proximity losses are computed independently against the same anchor and averaged; no term couples the two sensor modalities. We claim no novelty for the objective. Image–sensor and direct sensor–sensor alignment baselines are expressed in the same framework by weighting the corresponding alignment terms; cw denotes the weight of the direct tactile–proximity term ($cw = 0$: image anchor alone; $cw = 1$: with the direct pair).

C. Integration with Policy Learning

We use a Transformer-based diffusion policy [25], [28] consisting of an observation tokenizer, an observation encoder, and a noise-prediction network that generates action chunks by reversing the diffusion process conditioned on the observations. Only the sensor encoders g_m are transferred,

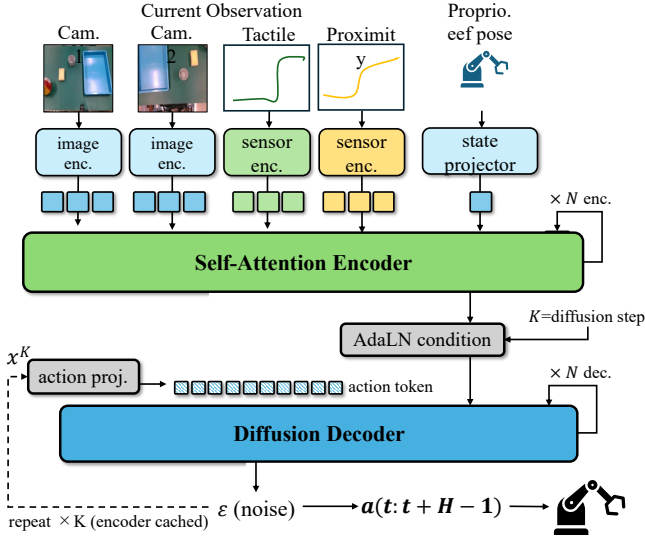


Fig. 3: Illustration of the model architecture for policy learning.

with the sensor ordering and history length L kept as in pretraining; they are fine-tuned end-to-end with the policy rather than frozen. Neither the anchor nor its encoder is transferred, so at inference the robot needs only images, proprioceptive state, and sensor histories.

V. EXPERIMENTAL SETUP

The robot is a UR5e with a Robotiq 2F-140 gripper carrying the fingertip module of Section III-A, which publishes three force and six proximity channels as ROS 2 topics at 100 Hz, independently of the 30 Hz policy control loop; the most recent samples are buffered and stored with the robot state and action at every control step. Two RealSense cameras (a wrist-mounted D435 and a workspace-mounted D415) provide a scene overview at 224×224 pixels. The robot state is a 6-D end-effector pose and a 2-D gripper opening; the action is seven commanded joint positions and one commanded gripper opening (Fig. 4).

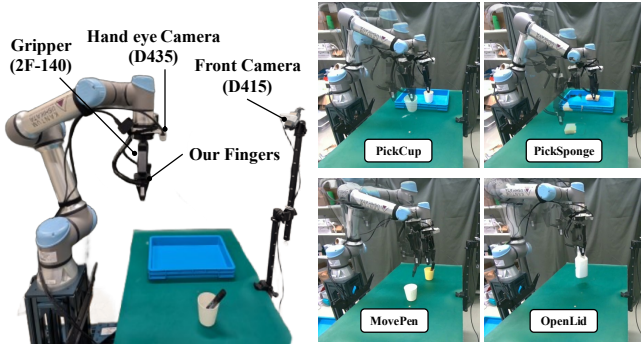


Fig. 4: Experimental Setup. Our fingertip can observe tactile and proximity signals, while 2 RealSense cameras observe the scene (one camera is visible above; another is mounted on the end-effector).

A. Data Collection

We evaluate our method on four real-robot manipulation tasks: PickCup, PickSponge, MovePen, and OpenLid. In PickCup and PickSponge, shown in Fig. 4, the robot grasps an object placed on a table and lifts it. In MovePen, the robot grasps a pen and transfers it into a container. In OpenLid, the robot grasps a container and opens its lid. For each task, we collect 50 demonstration episodes using a 3D mouse, randomizing the initial object position in every episode.

B. Model Training

Pretraining samples are obtained by subsampling the recorded sequence every five steps. Each sensor history x_t^m contains the latest $L = 50$ samples (0.5 s) of the 100 Hz sensor stream, read from the buffer at the corresponding control step so that it is aligned with the state and action of that step, while the action chunk $\bar{a}_t$ contains $H = 10$ subsampled steps, covering approximately 1.67 s at the 30 Hz control rate. The anchor dimension is $P + HA = 88$.

Mini-batches for the proposed condition are sampled non-uniformly: steps whose mean proximity value falls below the 20th percentile of its distribution, and steps at which the tactile amplitude crosses its 75th percentile in either direction, receive four times the sampling weight, and 25% of each batch is drawn from the same interval of other episodes as hard negatives. The thresholds are computed from the evaluated sensors but are not used in the anchor. Image-anchored conditions use uniform sampling, so the difference from the proposed condition includes this sampling rule as part of the training recipe.

The image-anchored baseline uses an ImageNet-1K-pretrained ResNet-18 with batch normalization replaced by 16-group normalization.

All models are trained with AdamW (learning rate 10^{-4} , weight decay 10^{-6} , batch size 512) for 1000 epochs without data augmentation; 96% of the episodes are used for training and the rest for validation, and the checkpoint with the lowest validation loss is selected. All conditions use the symmetric InfoNCE objective of Eq. (4) with a learnable temperature τ initialized to 0.07. Each condition is trained with three seeds on one Intel Xeon Platinum 8558 CPU and one NVIDIA H200 GPU (141 GB VRAM).

The policy (Fig. 3) takes two camera images, the robot state, and sensor histories constructed as in pretraining, and is trained for 1000 epochs with batch size 64, 100 diffusion training steps, 8 inference denoising steps, and an action horizon of 16. Each of three seeds is evaluated over 50 trials; we report the mean success rate and standard deviation.

C. Evaluation Metrics

Contact onset and intervals. The contact onset t_c is defined as the first step at which any tactile channel deviates from its resting value by more than 150 AD units. The approach interval covers the 60 control steps (2 s at 30 Hz) preceding t_c .

Representation analysis. Time-to-contact (TTC) is estimated from the frozen proximity embedding over the approach interval using ridge regression and reported as R^2 . Cross-sensor retrieval evaluates whether a tactile embedding retrieves the proximity embedding from the same time step, reported as Recall@1 normalized by chance.

VI. RESULTS AND DISCUSSION

We evaluate the proposed tactile–proximity sensing and contrastive pretraining for policy learning from limited demonstrations on four real-robot tasks, focusing on whether aligning sensor histories with robot-side signals that connect the current embodied state to future actions is effective. We consider the following two types of comparison conditions. All conditions use the same InfoNCE objective (Section V-B).

- 1) **Vision only / no pretraining:** A vision-only policy without fingertip sensing, and a policy with sensor histories concatenated to the observations without pretraining; the former isolates the effect of adding fingertip sensing, the latter the need for pretraining.
- 2) **Image anchor:** Following VITaL [3], each sensor representation is aligned with an image representation, optionally with an additional direct alignment term between same-time tactile and proximity representations (**image anchor + direct pair**, $cw = 1$).

Encoder architecture and training settings are shared across conditions; only the mini-batch sampling differs, as the proposed method uses the non-uniform sampling of Section V-B.

We evaluate manipulation success rate and the pretrained representations. For the latter, we ask whether the remaining time to contact (TTC) can be linearly decoded from the proximity representation in the pre-contact interval, where the tactile sensor has not yet responded, and report the coefficient of determination R^2 (1: perfect prediction; 0: predicting the mean), using raw sensor values and an untrained encoder as references (Section V-C). Pretraining takes about 3–4 hours and policy training about 8 hours.

We address the following research questions:

- **RQ1:** Is pretraining with robot-side signals as an anchor more effective than image-anchored pretraining (Table II)?
- **RQ2:** Can corresponding representations emerge between tactile and proximity sensing without directly aligning the two modalities during training (Fig. 5)?
- **RQ3:** Do the learned representations preserve pre-contact information, and does this information remain available for data collected during policy execution (Table III, Figs. 6 and 7)?

A. RQ1: Does anchoring to past, present, and future body signals help, compared with image anchors and direct pairs?

Without pretraining, directly concatenating sensor histories increases the average success rate from 30.4% to 40.4%, but performance on MovePen and OpenLid remains below that of the vision-only policy (Table I). With pretraining, the average success rate exceeds 70% for all conditions, and

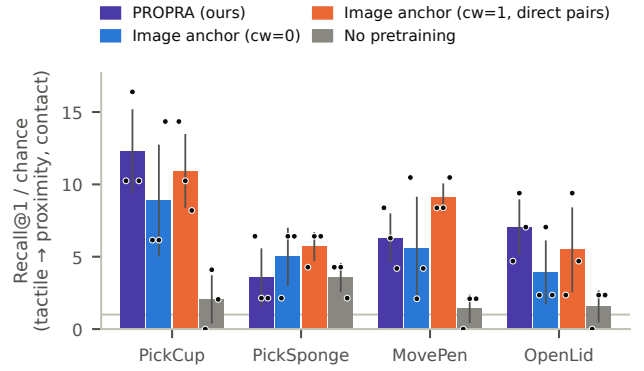


Fig. 5: Finding the proximity embedding of the same time step from the tactile embedding during contact, on episodes not used for training. Accuracy is divided by chance (line at 1). Only Image anchor + direct pair is trained on tactile–proximity pairs; No pretraining uses random weights. Bars: mean over three seeds; error bars: SD; dots: seeds. One correct retrieval changes a seed’s value by 2.0–2.4.

performance on these two tasks also surpasses the vision-only baseline (Table II).

PROPRA achieves the highest average success rate (75.0%, vs. 71.7% for the image anchor and 72.1% with the direct pair). It improves over the image anchor by 6.7 points on both PickCup and PickSponge (5.0 and 10.0 points over the direct-pair variant), although these differences are within the variation across seeds; MovePen is identical, and the image anchor is 3.4 points higher on OpenLid. Adding direct tactile–proximity alignment changes the average by only 0.4 points, so explicitly coupling the two representations brings no clear benefit.

Overall, robot-side anchoring matches or exceeds image-anchored pretraining on average without requiring camera images during pretraining.

B. RQ2: Do tactile and proximity representations correspond without direct pairing?

Although PROPRA never aligns tactile and proximity representations directly, we test whether their correspondence emerges by retrieving, for each tactile embedding during contact in held-out episodes, the proximity embedding of the same time step, reported relative to chance (Fig. 5).

On PickCup, MovePen, and OpenLid, PROPRA reaches 6–12× chance, above the untrained encoder (1–2×) for every seed combination, so correspondence can emerge indirectly through the shared anchor. On PickSponge, however, PROPRA reaches only 3.6×, comparable to the untrained encoder, and the image anchor with the direct pair exceeds PROPRA on PickSponge and MovePen.

C. RQ3: Does the representation retain pre-contact information, also beyond the demonstration data?

Over the 2 s before contact (Table III, Fig. 6), all pretrained representations predict TTC more accurately than the raw signal, and PROPRA gives the highest R^2 on three of the

TABLE II: Effect of pretraining and anchor choice with both fingertip sensors. Success rate [%], mean $\pm$ SD over 3 seeds with 50 trials per seed. + direct pair adds direct tactile–proximity alignment. Bold and underlined values indicate the best and second-best results in each column, respectively.

Pretraining	PickCup	PickSponge	MovePen	OpenLid	Avg.
None	61.7 $\pm$ 5.8	51.7 $\pm$ 12.6	20.0 $\pm$ 8.7	28.3 $\pm$ 15.3	40.4
Image anchor	75.0 $\pm$ 13.2	<u>75.0</u> $\pm$ 13.2	<u>60.0</u> $\pm$ 14.1	76.7 $\pm$ 12.6	71.7
Image anchor + pair	<u>76.7</u> $\pm$ 7.6	71.7 $\pm$ 2.9	63.3 $\pm$ 2.9	76.7 $\pm$ 7.6	<u>72.1</u>
PROPRA (ours)	81.7 $\pm$ 5.8	81.7 $\pm$ 7.6	63.3 $\pm$ 5.8	<u>73.3</u> $\pm$ 5.8	75.0

TABLE III: Time-to-contact estimation before contact: R^2 ($\uparrow$) from the frozen proximity embedding, mean $\pm$ SD over 3 seeds. No pretr.: random weights. Raw: sensor values. Bold and underlined values indicate the best and second-best pretrained condition.

Task	PROPRA	Image	Image + pair	No pretr.	Raw
PickCup	0.694 $\pm$ 0.002	<u>0.675</u> $\pm$ 0.011	0.664 $\pm$ 0.005	0.535 $\pm$ 0.010	0.511
PickSponge	0.440 $\pm$ 0.015	<u>0.377</u> $\pm$ 0.034	0.372 $\pm$ 0.035	0.145 $\pm$ 0.016	0.327
MovePen	0.544 $\pm$ 0.008	<u>0.545</u> $\pm$ 0.010	0.554 $\pm$ 0.007	0.500 $\pm$ 0.015	0.347
OpenLid	0.680 $\pm$ 0.026	<u>0.652</u> $\pm$ 0.006	0.643 $\pm$ 0.005	0.622 $\pm$ 0.009	0.400

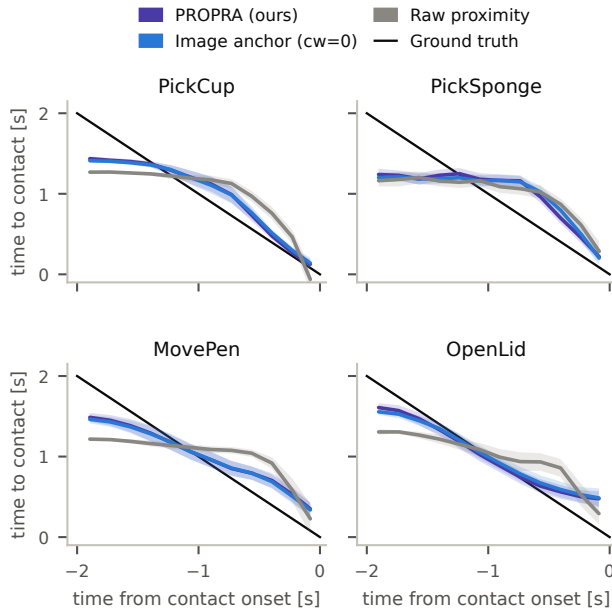


Fig. 6: Time to contact estimated from the frozen proximity embedding during the 2s before contact, averaged over the demonstration episodes (and over three seeds for the embeddings). Bands: 95% CI across episodes. The diagonal is the true value.

four tasks; the largest gain over the image anchor is on PickSponge, where prediction from the raw signal is hardest (0.440 vs. 0.377), while the image anchor with the direct pair is highest on MovePen. On MovePen and OpenLid, most of the gain over the raw signal is already obtained without pretraining (0.15 of 0.20 and 0.22 of 0.28), so the effect of pretraining is concentrated on PickCup and PickSponge. The

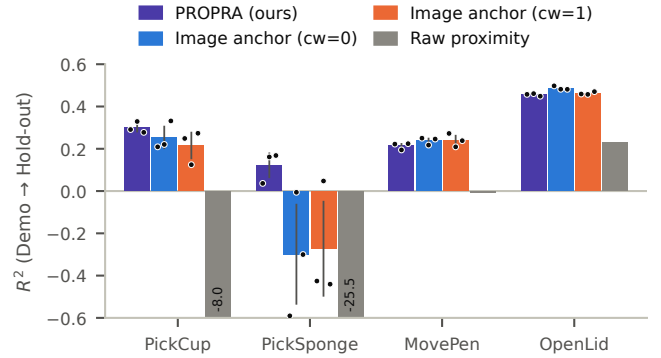


Fig. 7: R^2 of the time-to-contact estimate on new recordings from policy execution, using the regression fitted on the demonstrations without refitting. Bars: mean over three seeds; error bars: SD; dots: seeds. Values below the axis are printed inside the bars.

tactile representation does not predict TTC ($R^2 \leq 0.02$).

On 100 additional episodes per task recorded during policy execution and used for neither pretraining nor policy learning, after aligning each episode’s initial stationary proximity value with the demonstrations to compensate for zero-point drift, a linear predictor refit on these data retains 65–90% of PROPRA’s demonstration R^2 (95% CI within $\pm 4\%$), and PROPRA outperforms both image-anchor conditions in 20 of 24 seed-wise comparisons.

Applying the demonstration-fitted predictor without refitting (Fig. 7), the raw signal yields strongly negative R^2 on PickCup and PickSponge (-8.0 and -25.5), and both image-anchor conditions turn negative on PickSponge, whereas PROPRA stays positive on every task and seed; the image-anchor conditions are higher on MovePen and OpenLid.

D. Limitations and future perspective

Robot proprioception and actions require no additional data collection or annotation in this setting, a practical advantage of sensorimotor anchors, especially for sensors designed to capture phase changes during grasping. The proposed anchor is not uniformly superior, however: on OpenLid the image anchor reaches 76.7% versus 73.3% for PROPRA. Repeated contact while rotating the lid produces substantial fingertip variation over a long portion of the trajectory, so the sparse-response assumption is less pronounced and the demonstrations may already contain enough correspondence

between visual observations and sensor changes for image-based alignment. This is consistent with the advantage of PROPRA being most apparent when informative responses are temporally sparse, and it also shows that the anchor does not necessarily help when sensor information is rich over much of the trajectory, which remains an important limitation.

Our fingertip design and pretraining target a specific setting: grasp-oriented manipulation with two sensors retrofitted to an existing gripper and imitation learning from limited demonstrations. We therefore do not compare with large-scale foundation models built for particular tactile devices, since our focus is the sensing configuration and the pre-training objective. Combining PROPRA with multimodal foundation models such as Sparsh-X [12], for example as an adaptation stage when task-specific or non-standard sensors are introduced, and evaluating its scalability to broader sensing configurations, are left for future work.

VII. CONCLUSION

This study addresses the difficulty of obtaining the contact states needed for grasping from either external cameras or a single fingertip sensor. We built a gripper with proximity and tactile sensors that are informative before and after contact, respectively, and proposed PROPRA for pretraining their representations. The results indicate that sensorimotor signals serve as useful anchors for sensors whose informative responses are confined to particular intervals, without direct alignment between observation modalities. Future work will consider other tools and sensing modalities, including grasp force, torque, contact sound, and vibration, and test whether the learned representations remain effective across sensors with different physical characteristics and response patterns.

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